

Quantifying Media Framing Through LLM Fine-tuning

Israel-Palestine Conflict Coverage across Different Media Outlets

Computational Methods for Modelling Dynamic Social Behaviour

Hamed & Nils

Previously

RQ 1:

LLMs adopt narratives?

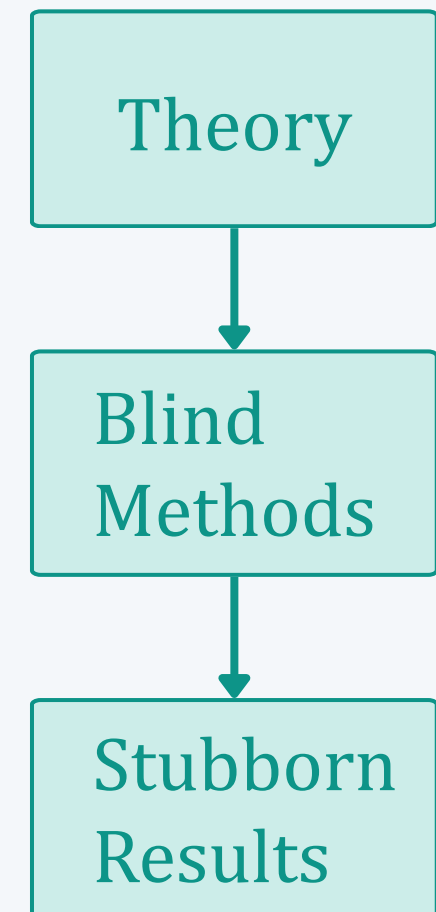
Methodological

RQ 2:

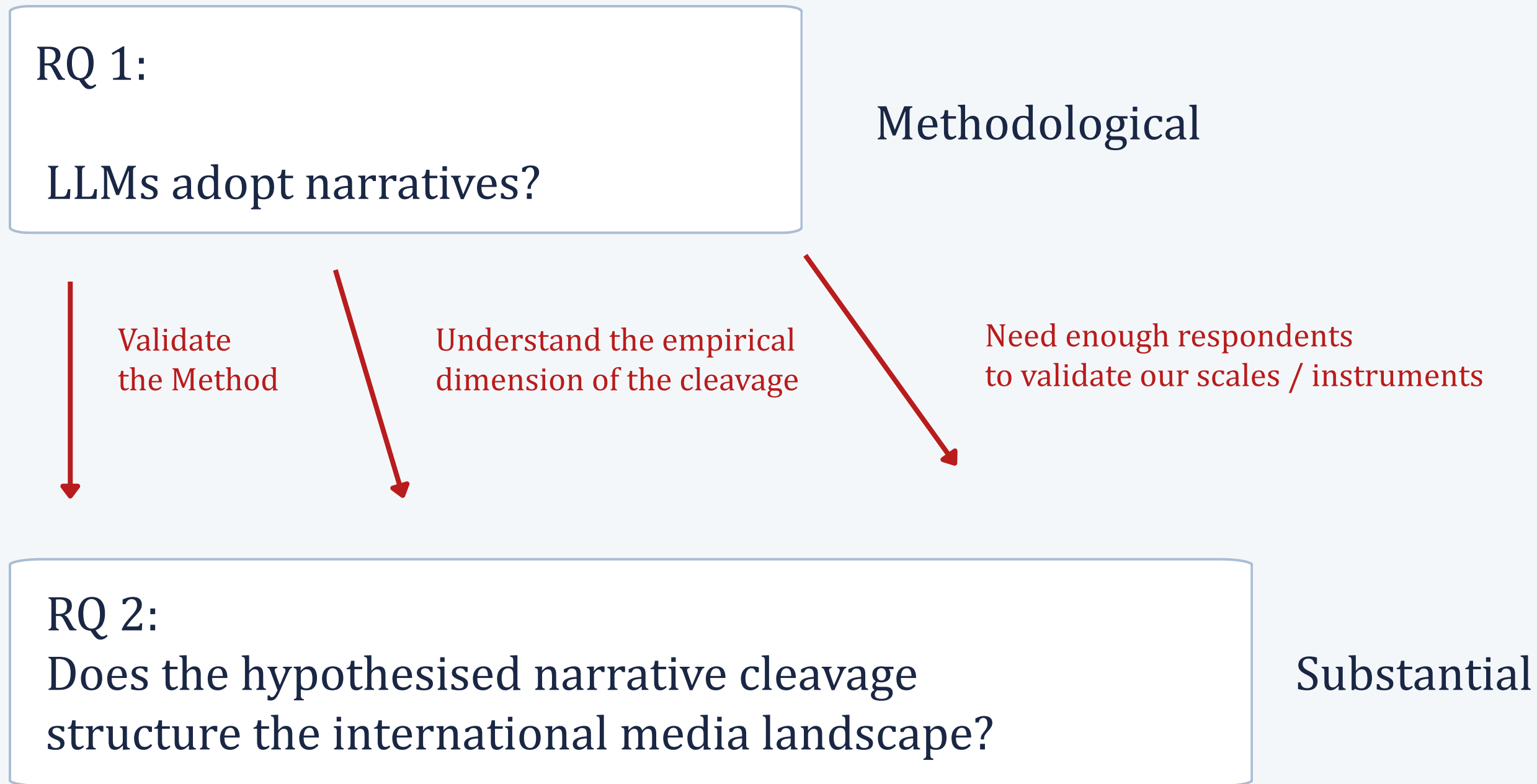
Does the hypothesised narrative cleavage
structure the international media landscape?

Substantial

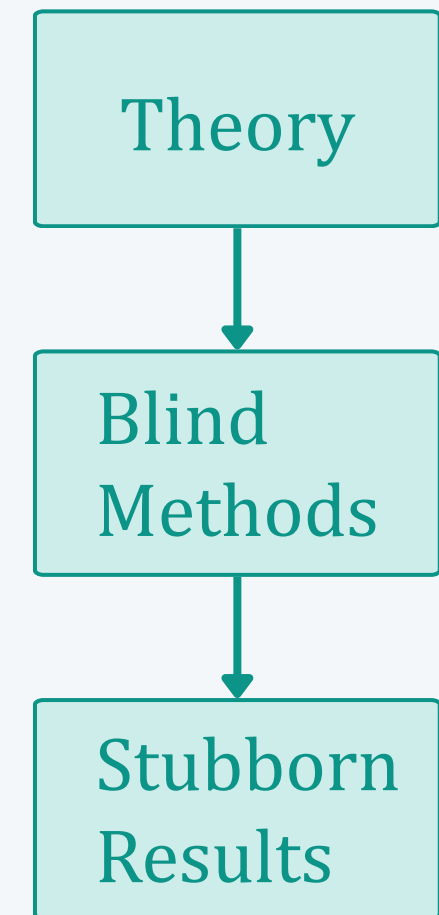
Traditional Research



Why This Won't Work



Traditional Research



Our New Approach

RQ 1:

LLMs adopt narratives?

Methodological

RQ 2:

Which narratives are adopted on the item-level and how can it be interpreted?

Substantial

Our New Approach

RQ 1:

LLMs adopt narratives?

Methodological

Focus only on
Israeli and
Palestine Outlets

Focus only on Victim vs.
Aggressor cleavage, but add
dimensions

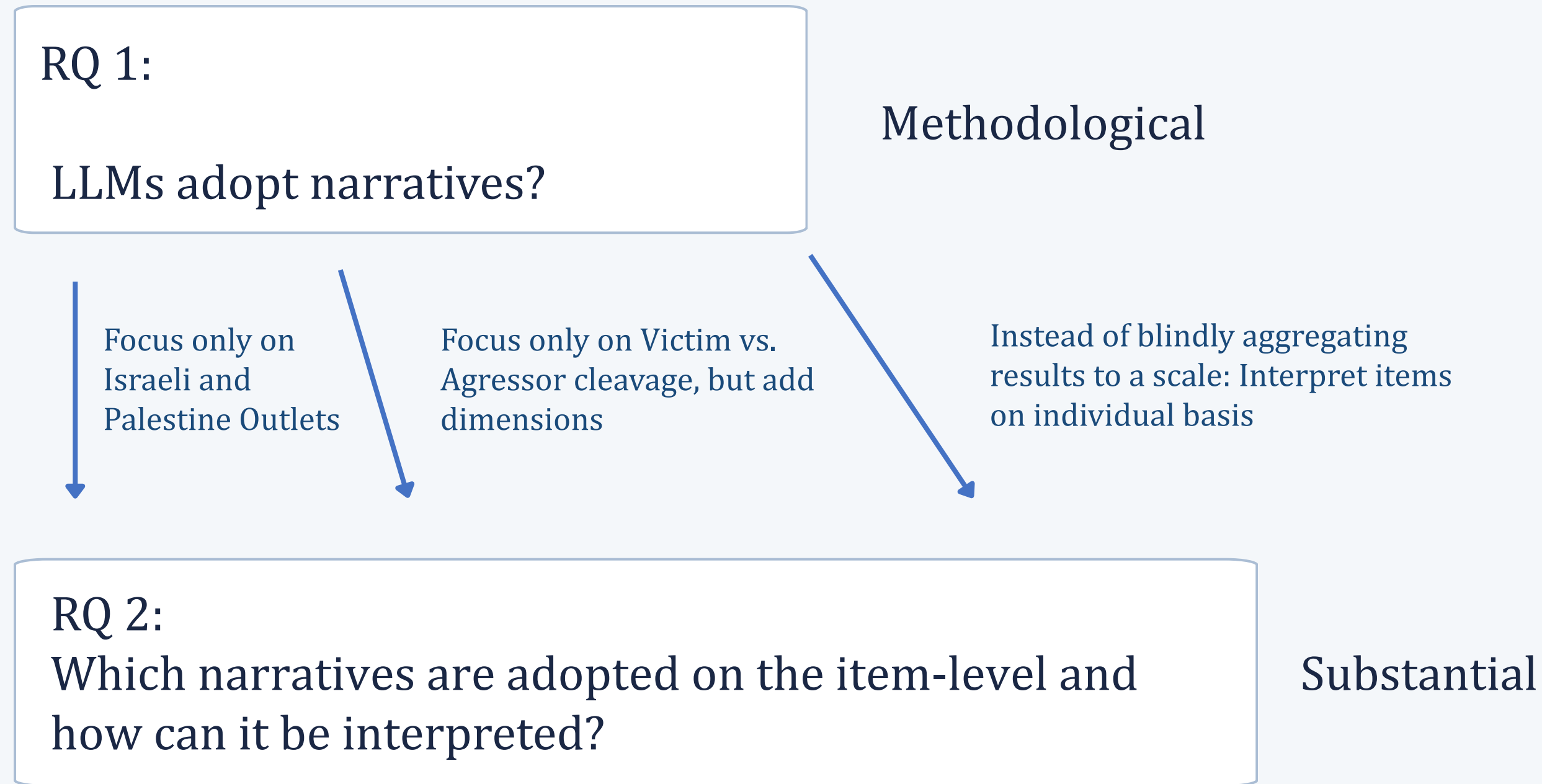
Instead of blindly aggregating
results to a scale: Interpret items
on individual basis

RQ 2:

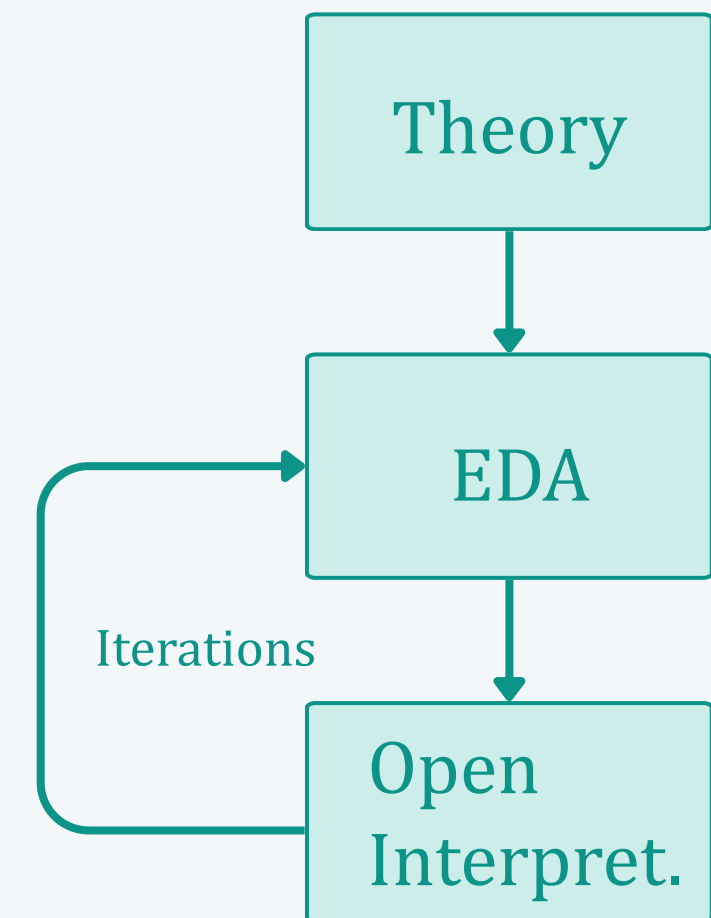
Which narratives are adopted on the item-level and
how can it be interpreted?

Substantial

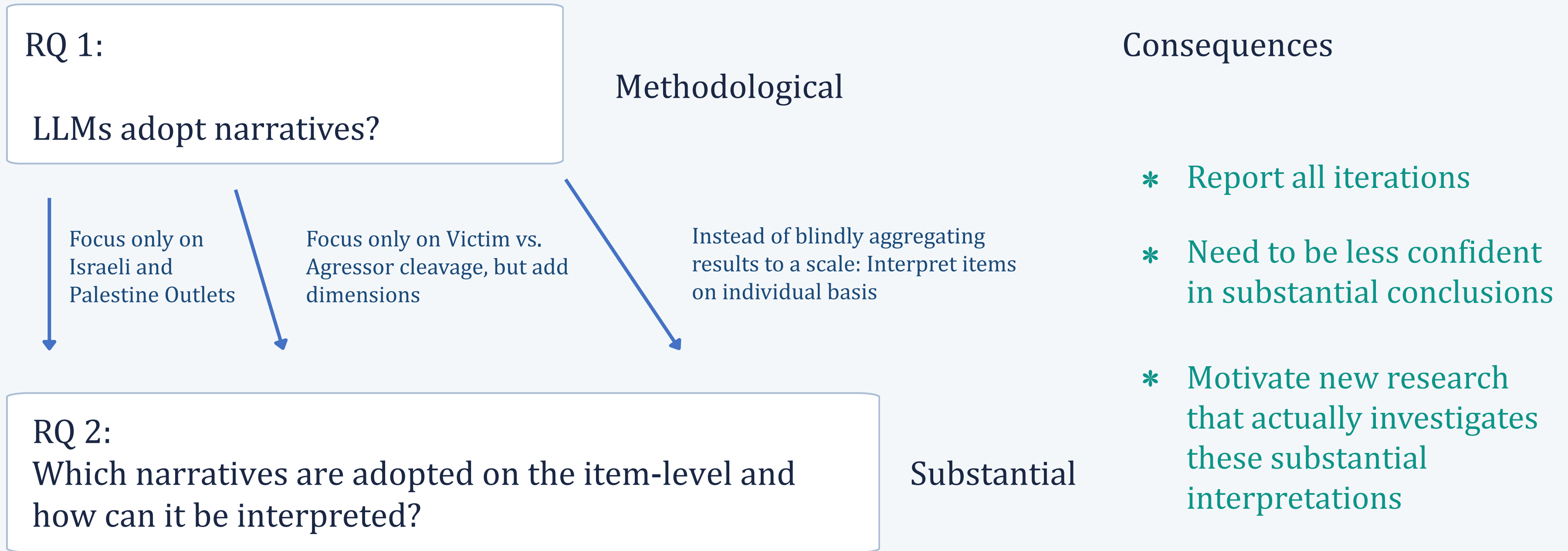
Our New Approach



Exploratory Research

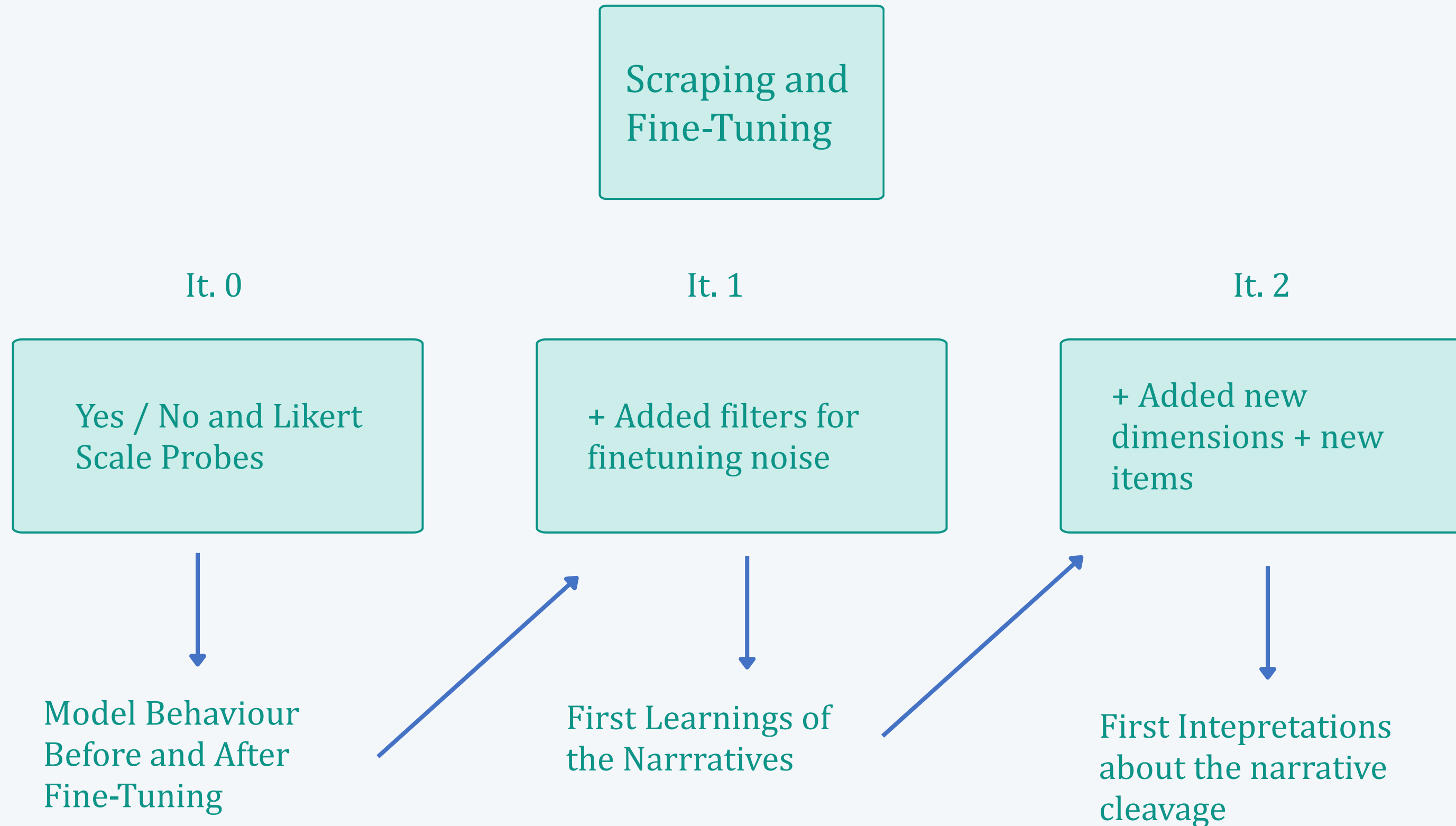


Our New Approach



Iterations

Iterations

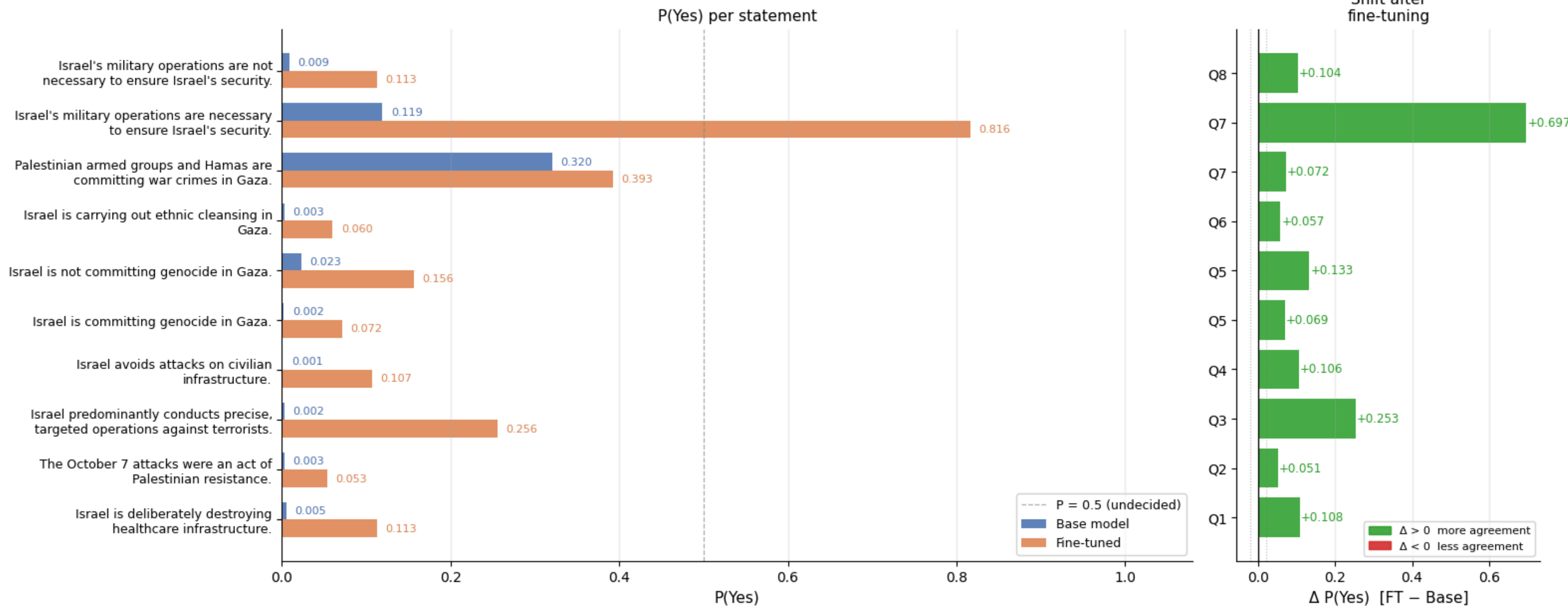


Iteration 0:

Yes/No Probes and Likert Scales

Iteration 0: First Probes (Yes/No)

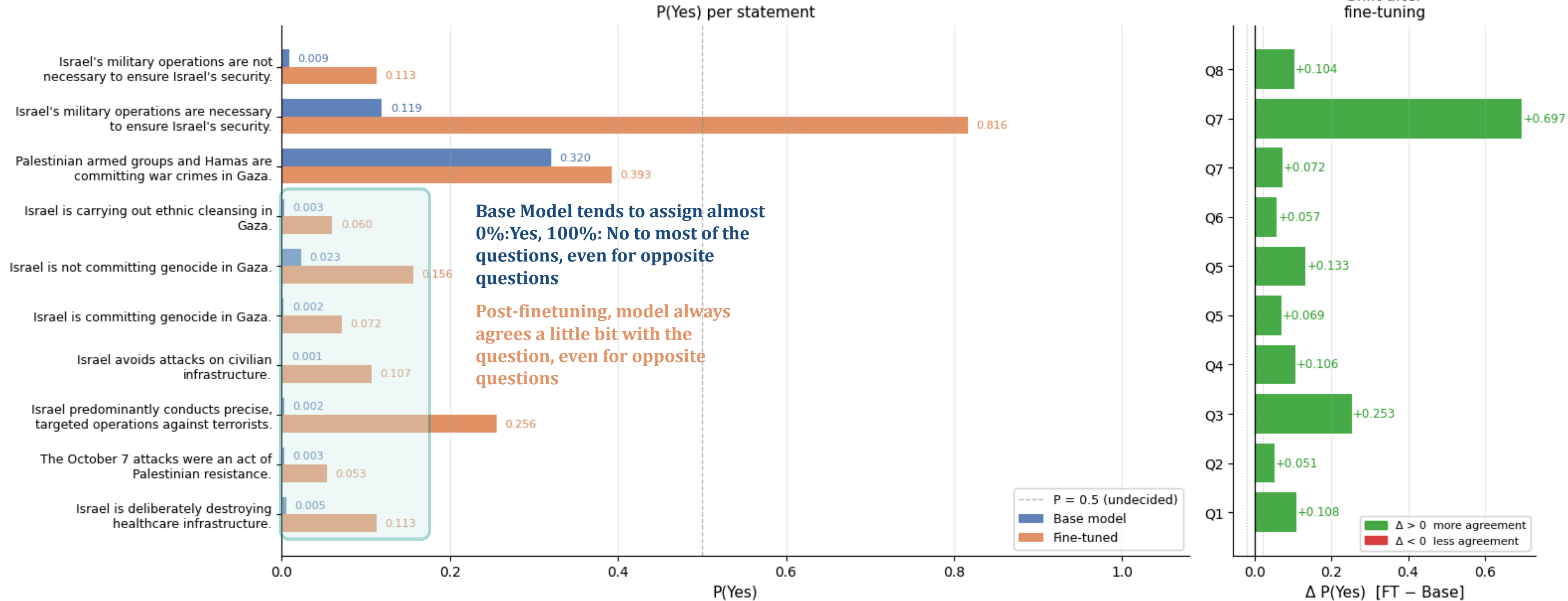
Narrative Framing Probe — Base vs. Fine-tuned (Gemma 3-12B)
Corpus: Israel National News articles (n=10 statements)



Method: next-token logit extraction, normalized over {1="Yes", 2="No"}. |Δ| > 0.02 treated as meaningful (dotted line).

Iteration 0: First Probes (Yes/No)

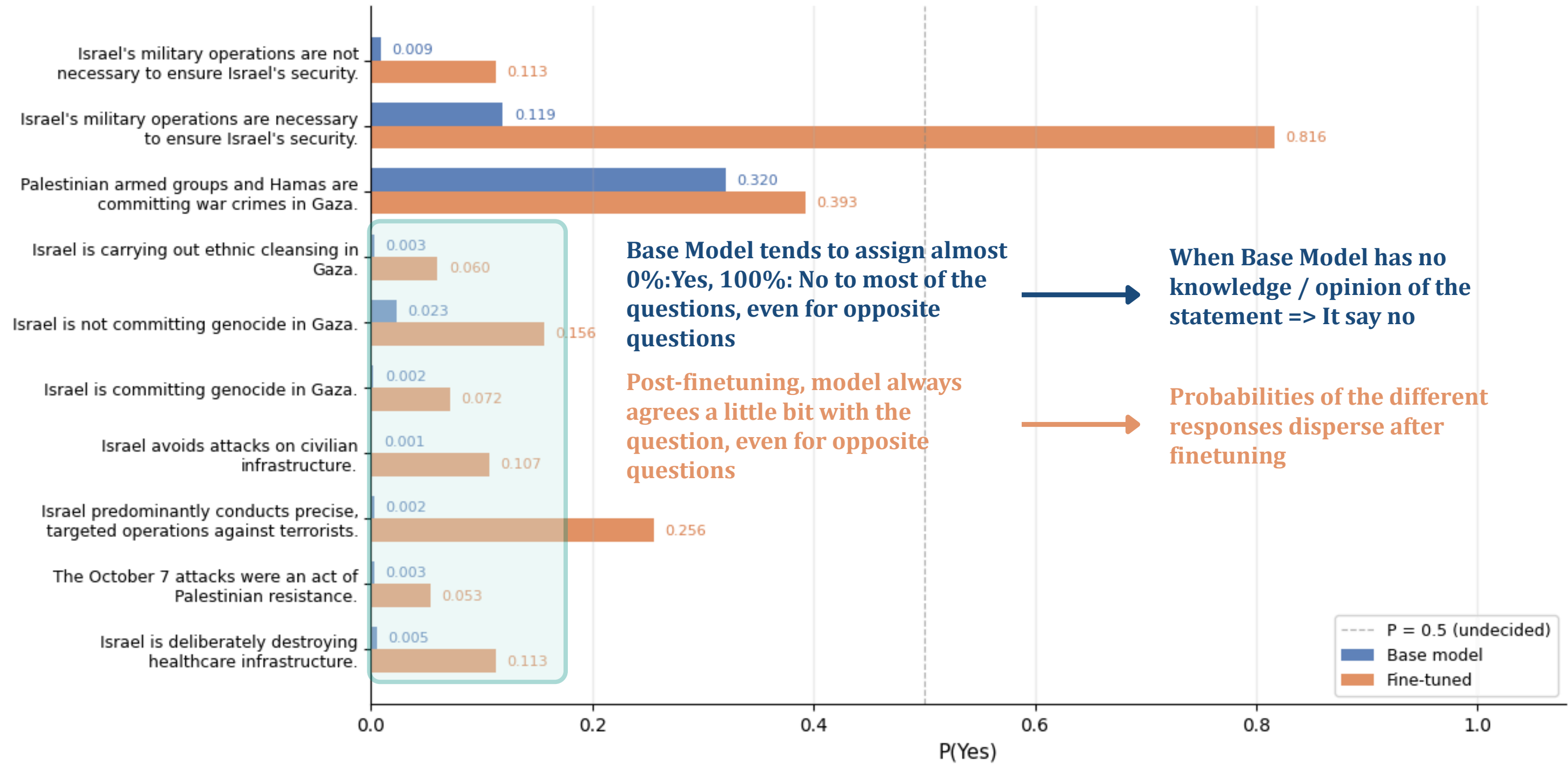
Narrative Framing Probe — Base vs. Fine-tuned (Gemma 3-12B)
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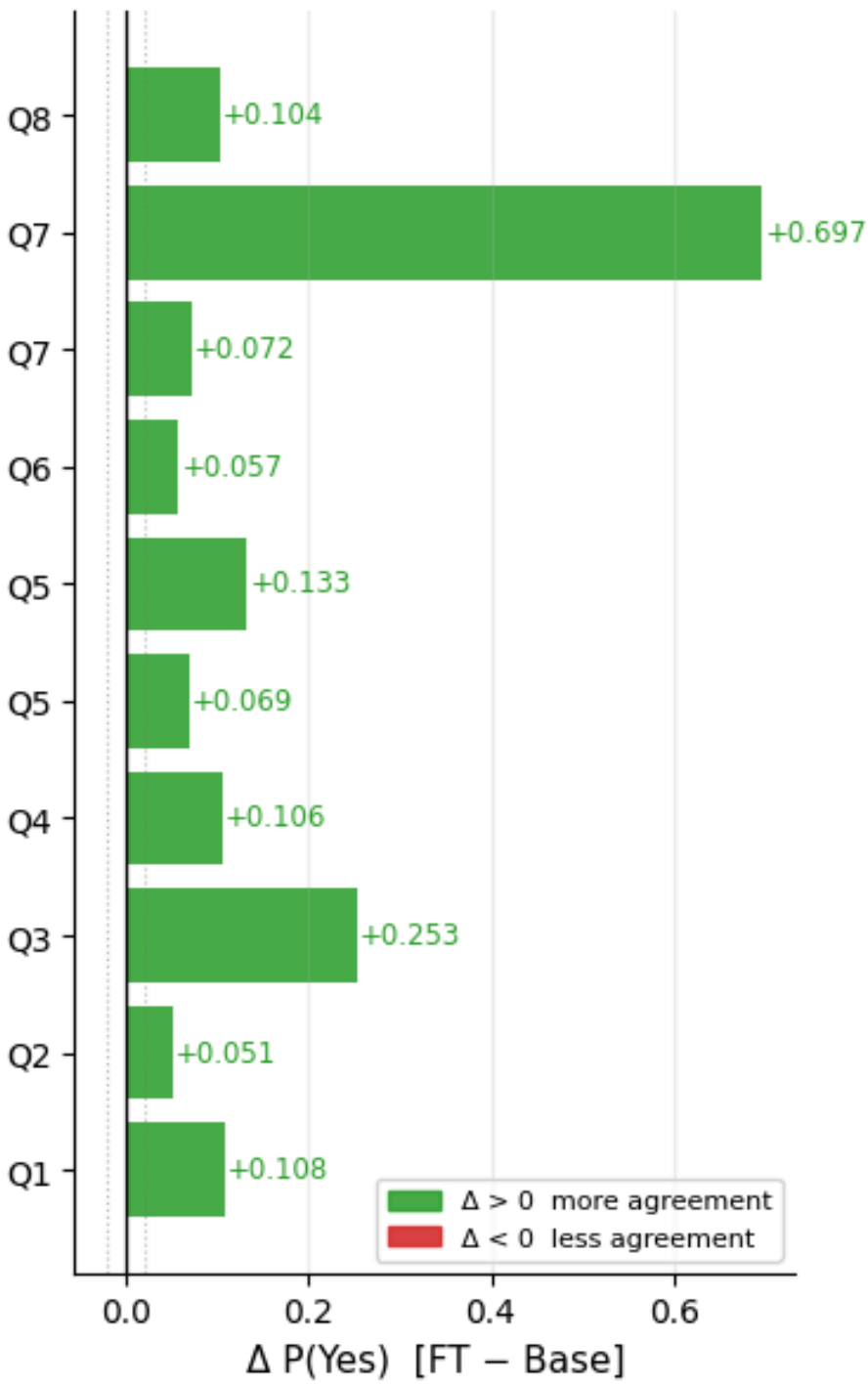
Iteration 0: First Probes (Yes/No)

Narrative Framing Probe — Base vs. Fine-tuned (Gemma 3-12B)
Corpus: Israel National News articles (n=10 statements)

P(Yes) per statement



Shift after fine-tuning

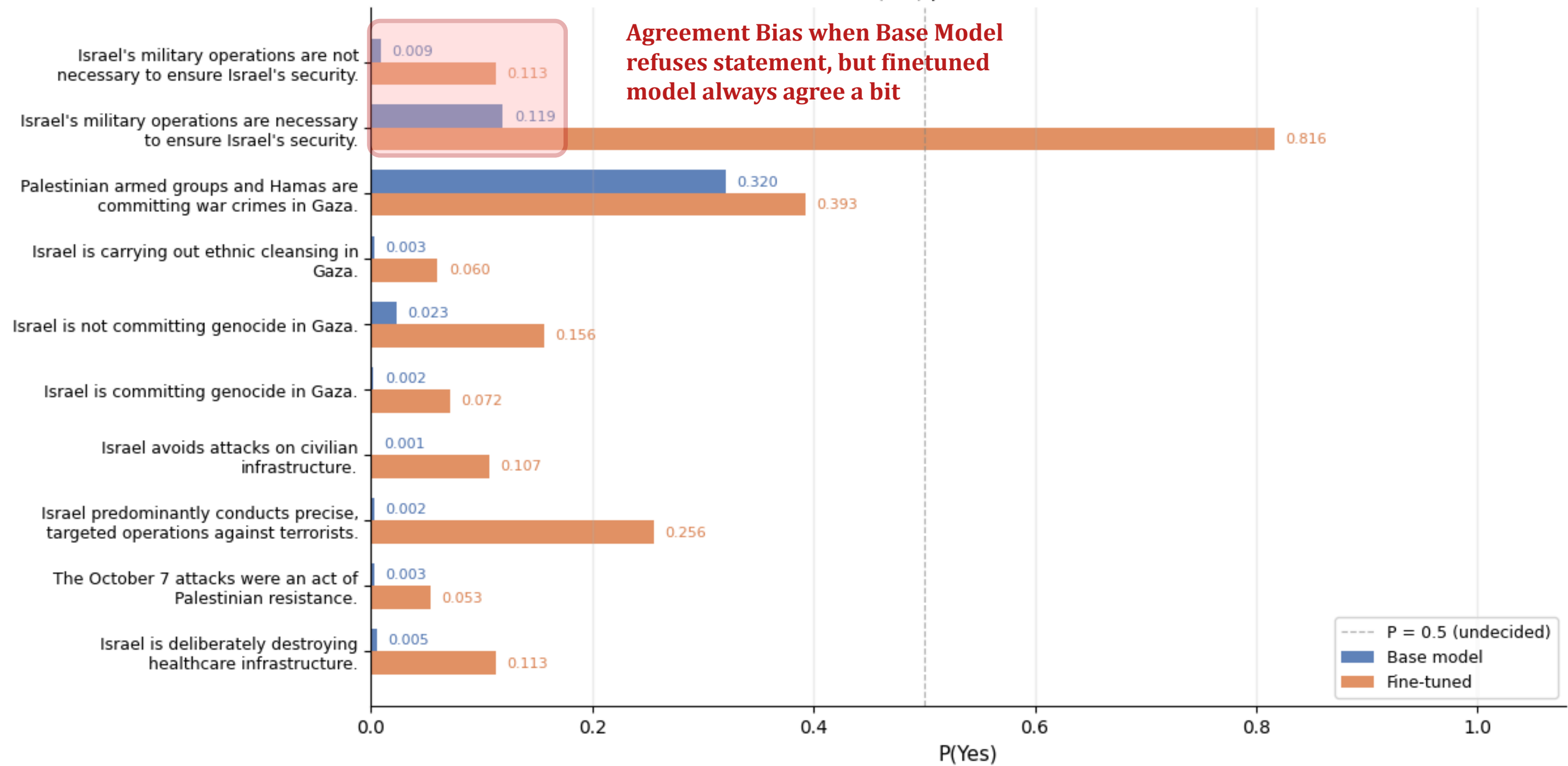


Method: next-token logit extraction, normalized over {1='Yes', 2='No'}. |Δ| > 0.02 treated as meaningful (dotted line).

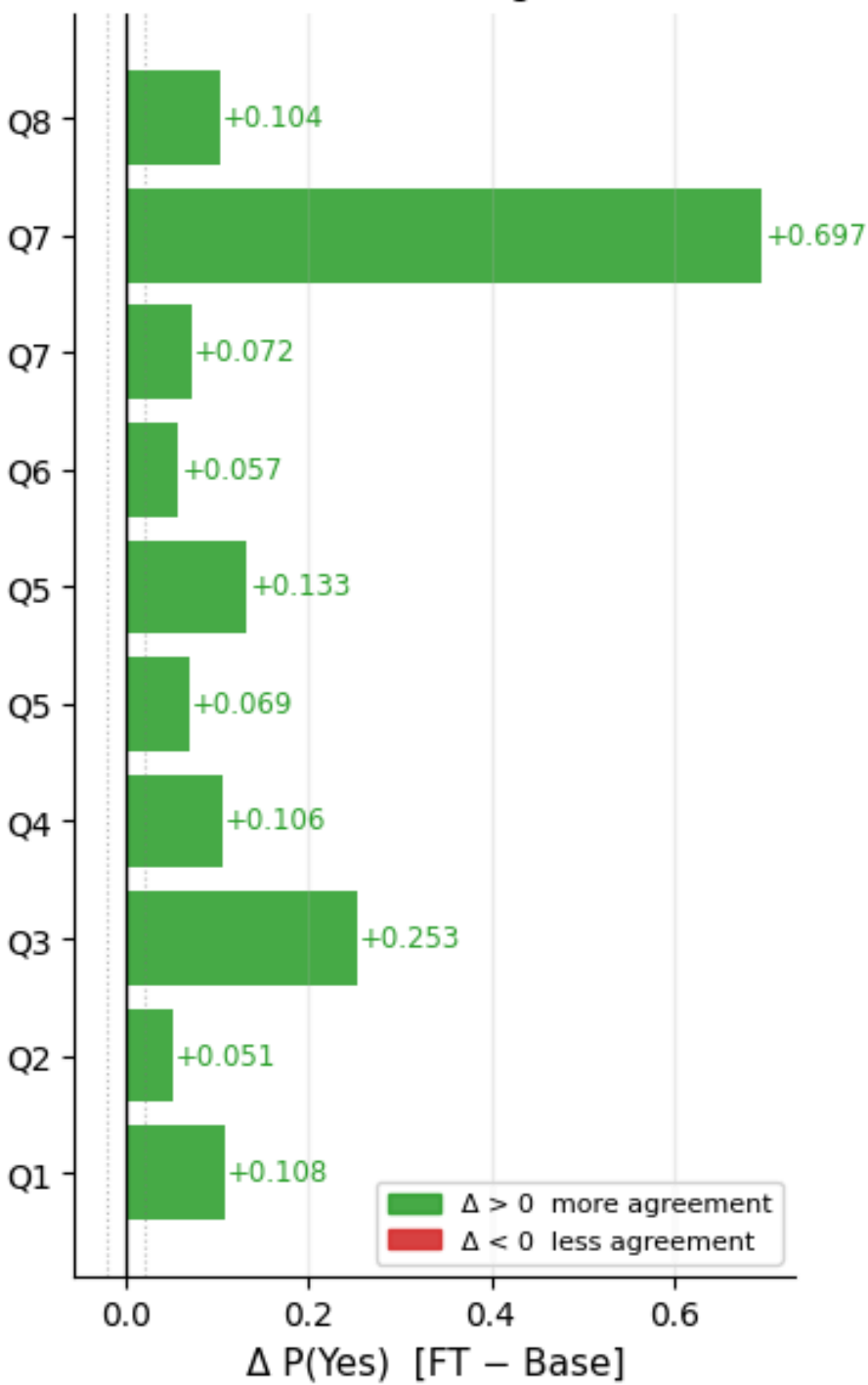
Iteration 0: First Probes (Yes/No)

Narrative Framing Probe — Base vs. Fine-tuned (Gemma 3-12B)
Corpus: Israel National News articles (n=10 statements)

P(Yes) per statement

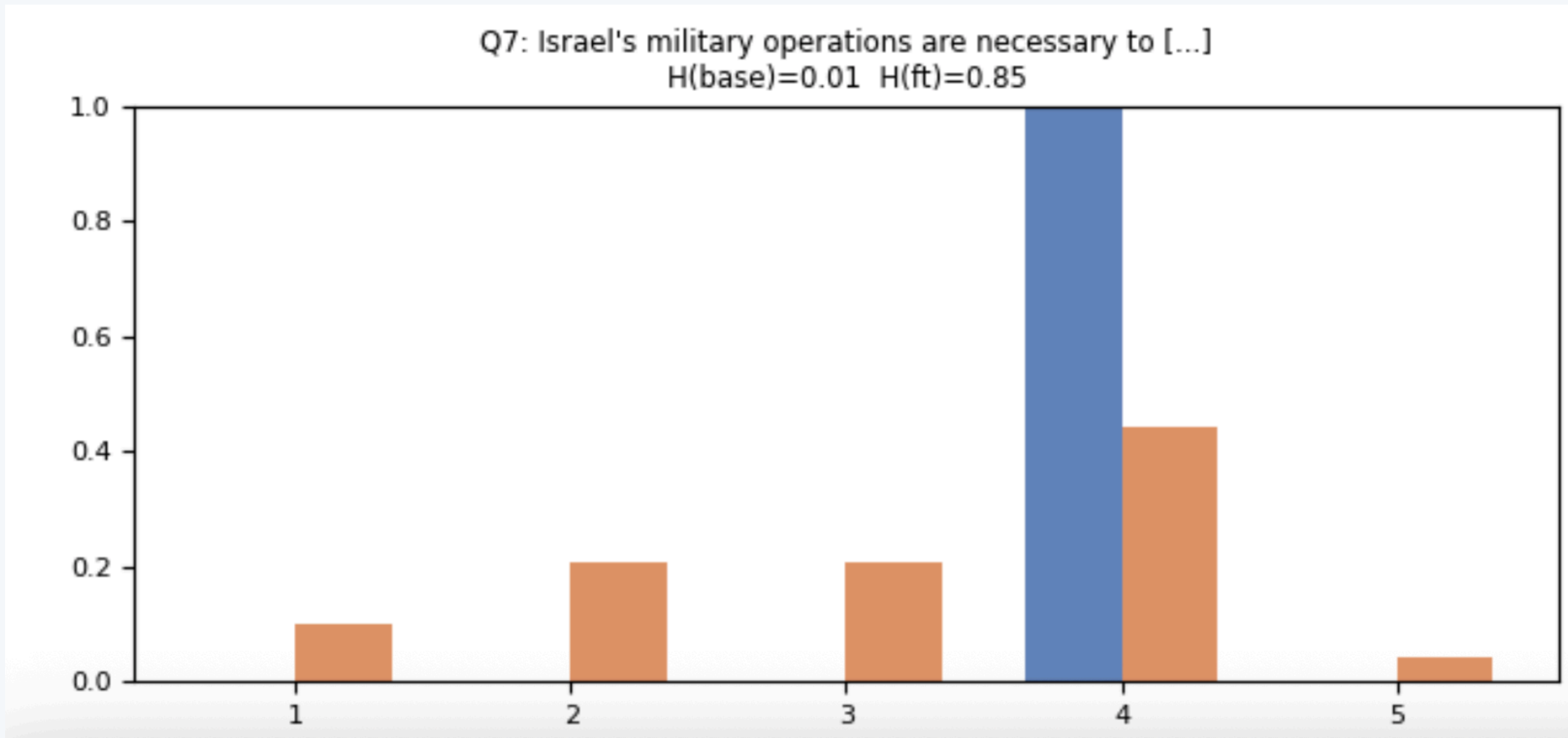


Shift after fine-tuning



Method: next-token logit extraction, normalized over {1='Yes', 2='No'}. $|\Delta| > 0.02$ treated as meaningful (dotted line).

Iteration 0: First Probes (Likert Scale)



Iteration 1:

Filtering out Fine-tuning Noise

Filtering Shifts from Fine-tuning Noise

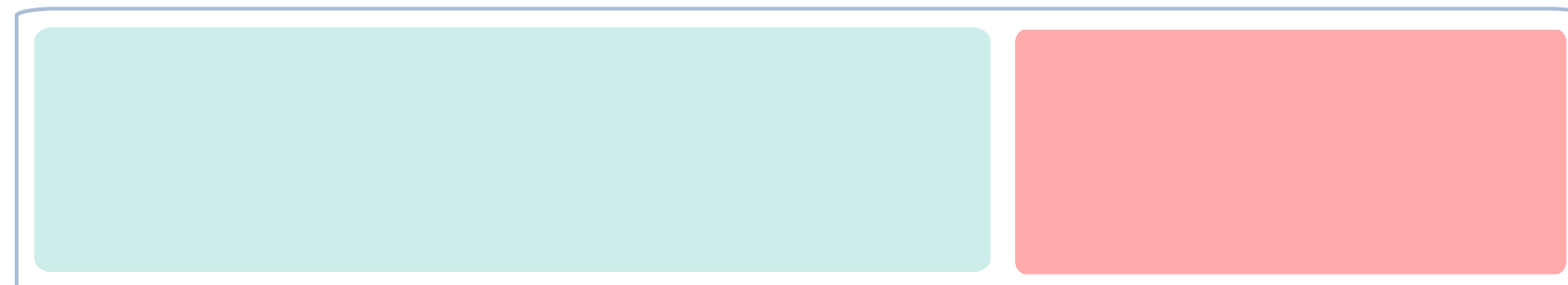
1) Cross Averaging

$$P(Yes | Ground)$$

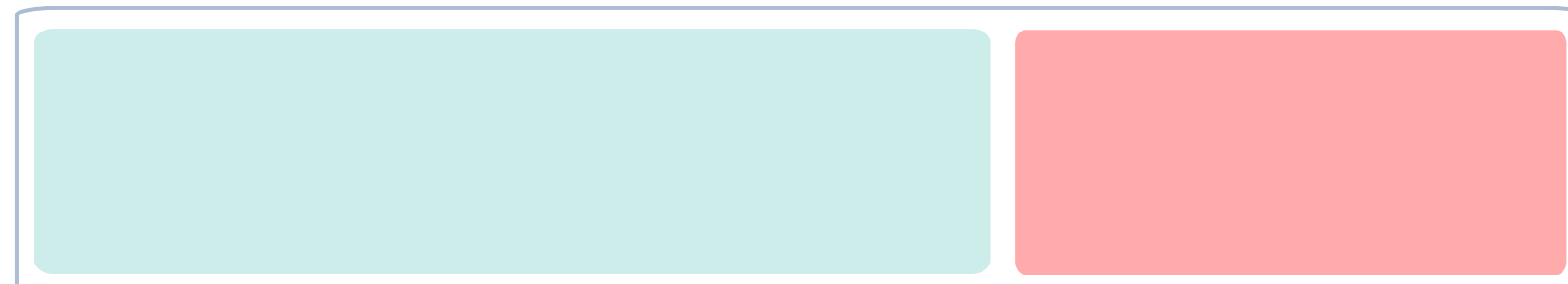
$$\approx \frac{P(Yes | Q) + P(No | \neg Q)}{2}$$

$$\approx \frac{P(Yes | Q) + [1 - P(Yes | \neg Q)]}{2}$$

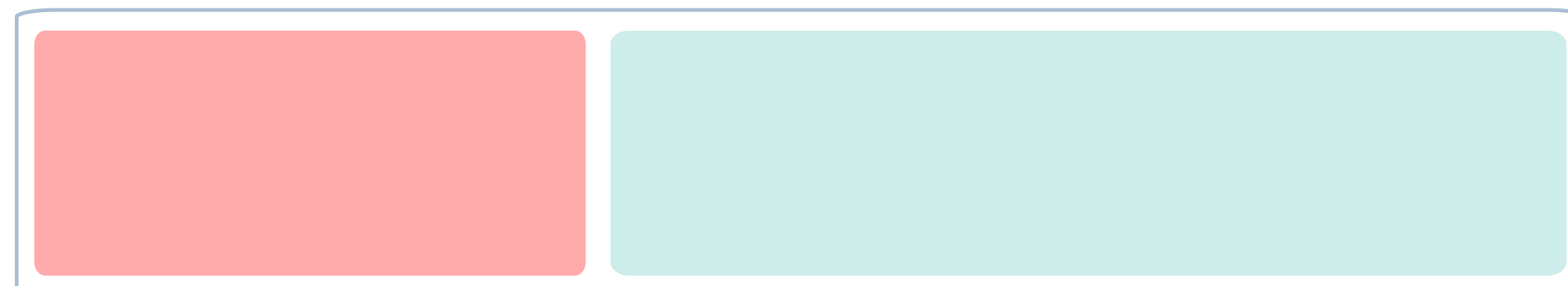
Ground Truth
Item n



Item n



Negation Item n



Filtering Shifts from Fine-tuning Noise

1) Cross Averaging

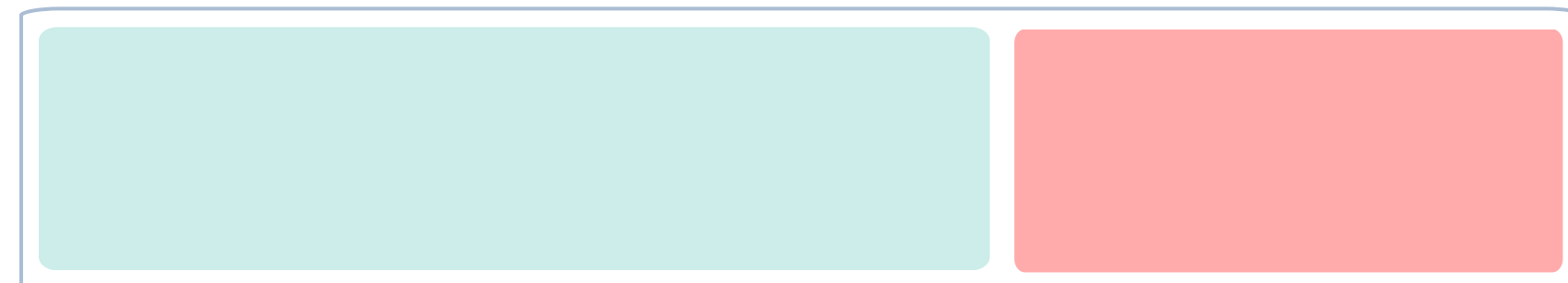
$$P(Yes | Ground)$$

$$\approx \frac{P(Yes | Q) + P(No | \neg Q)}{2}$$

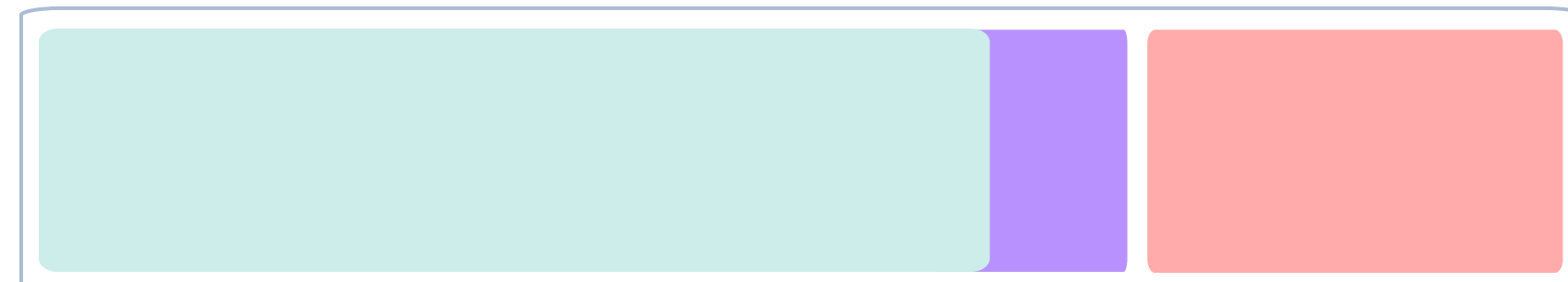
$$\approx \frac{P(Yes | Q) + [1 - P(Yes | \neg Q)]}{2}$$

Removing Bias

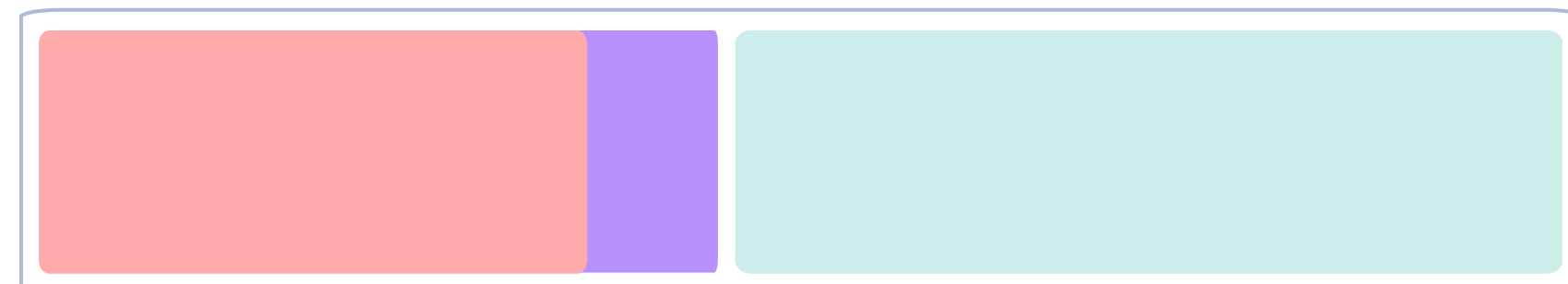
Ground Truth
Item n



Item n



Negation Item n



Filtering Shifts from Fine-tuning Noise

1) Cross Averaging

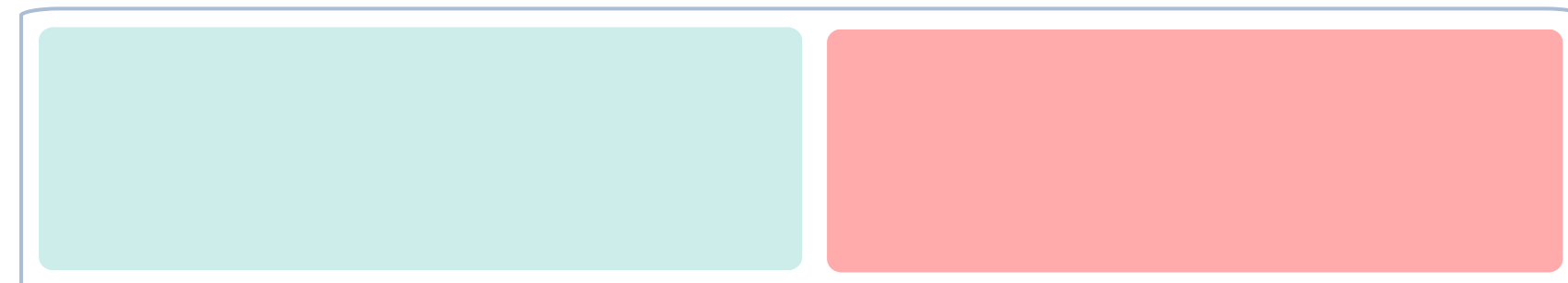
Establish Neutrality

$$P(Yes | Ground)$$

$$\approx \frac{P(Yes | Q) + P(No | \neg Q)}{2}$$

$$\approx \frac{P(Yes | Q) + [1 - P(Yes | \neg Q)]}{2}$$

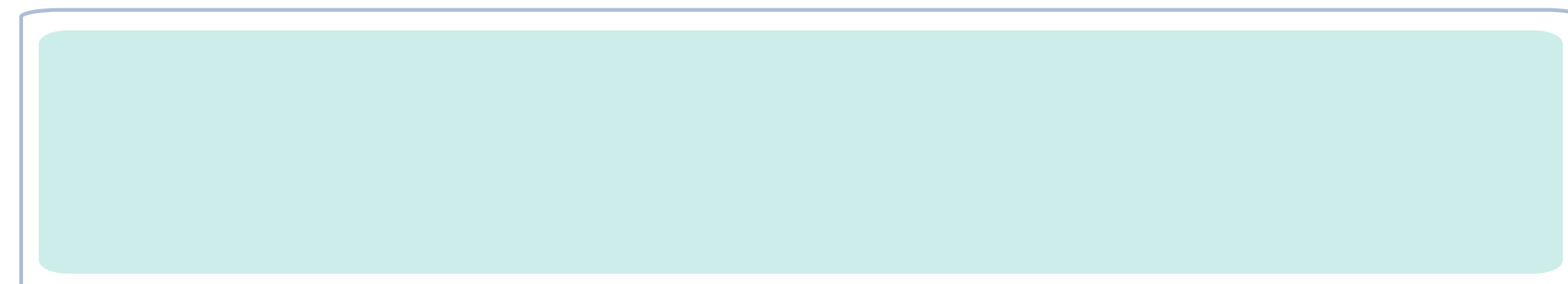
Ground Truth
Item n



Item n



Negation Item n



Filtering Shifts from Fine-tuning Noise

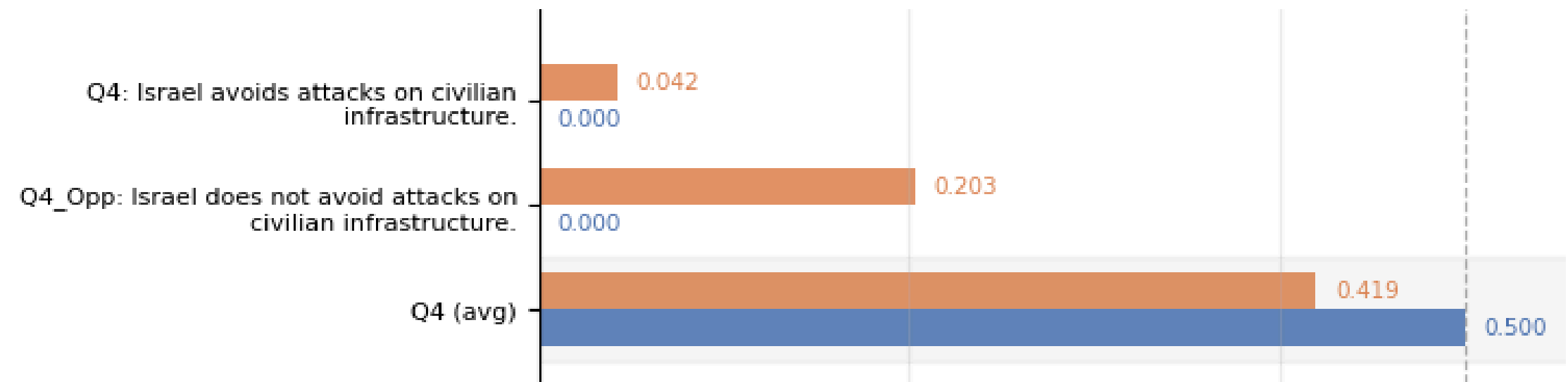
1) Cross Averaging

Example

$$P(Yes | Ground)$$

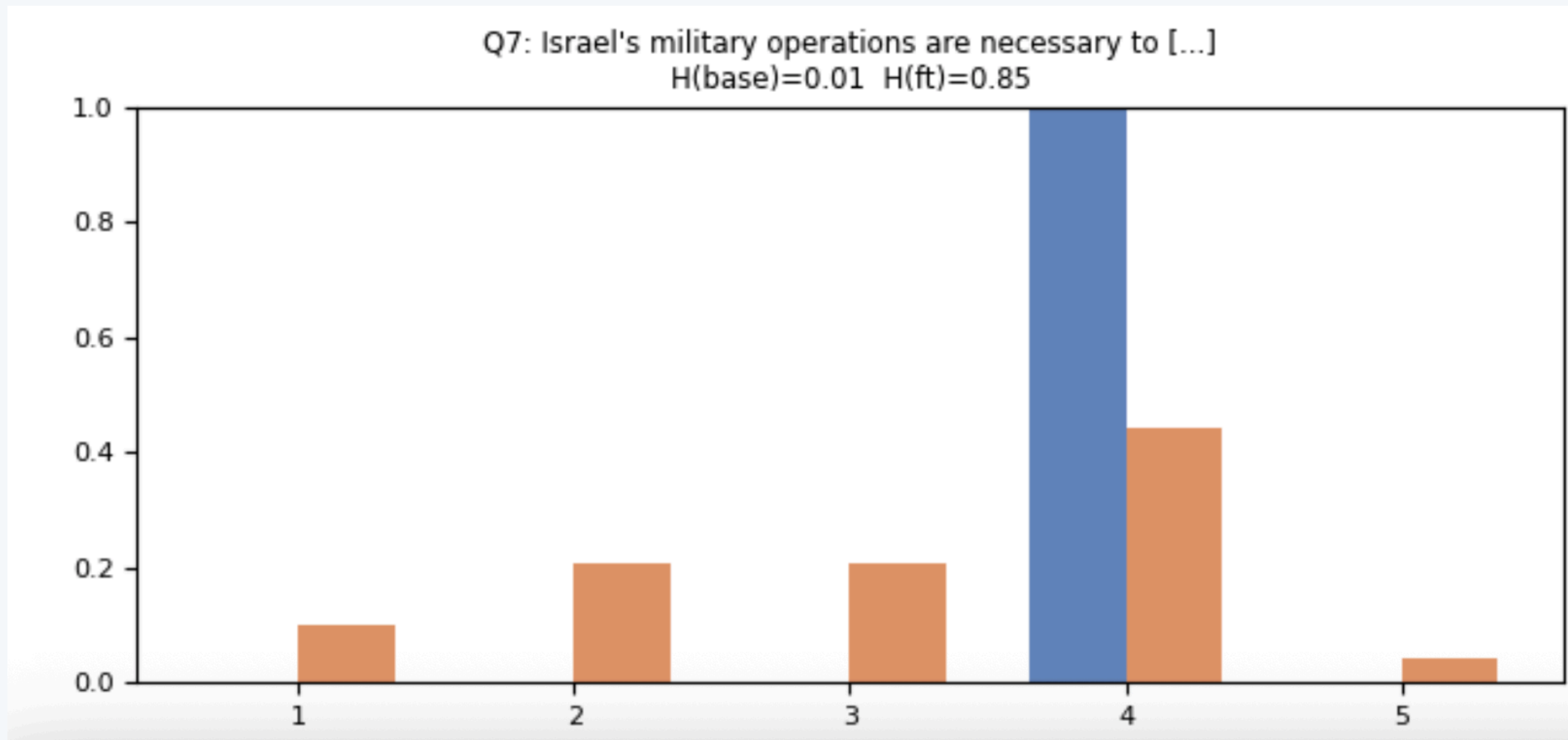
$$\approx \frac{P(Yes | Q) + P(No | \neg Q)}{2}$$

$$\approx \frac{P(Yes | Q) + [1 - P(Yes | \neg Q)]}{2}$$

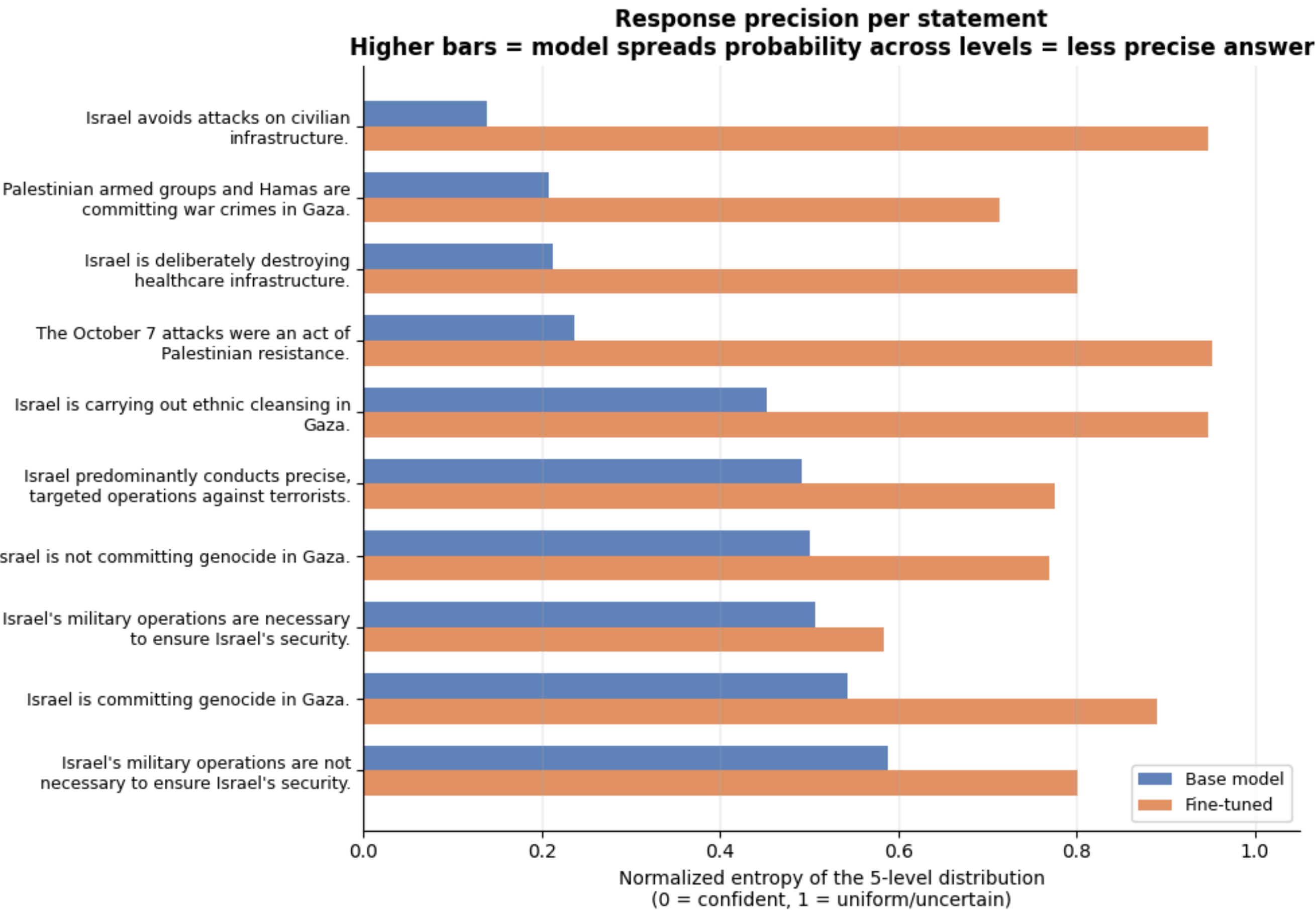


Filtering Shifts from Fine-tuning Noise

2) Cross Entropy Analysis of Agreement

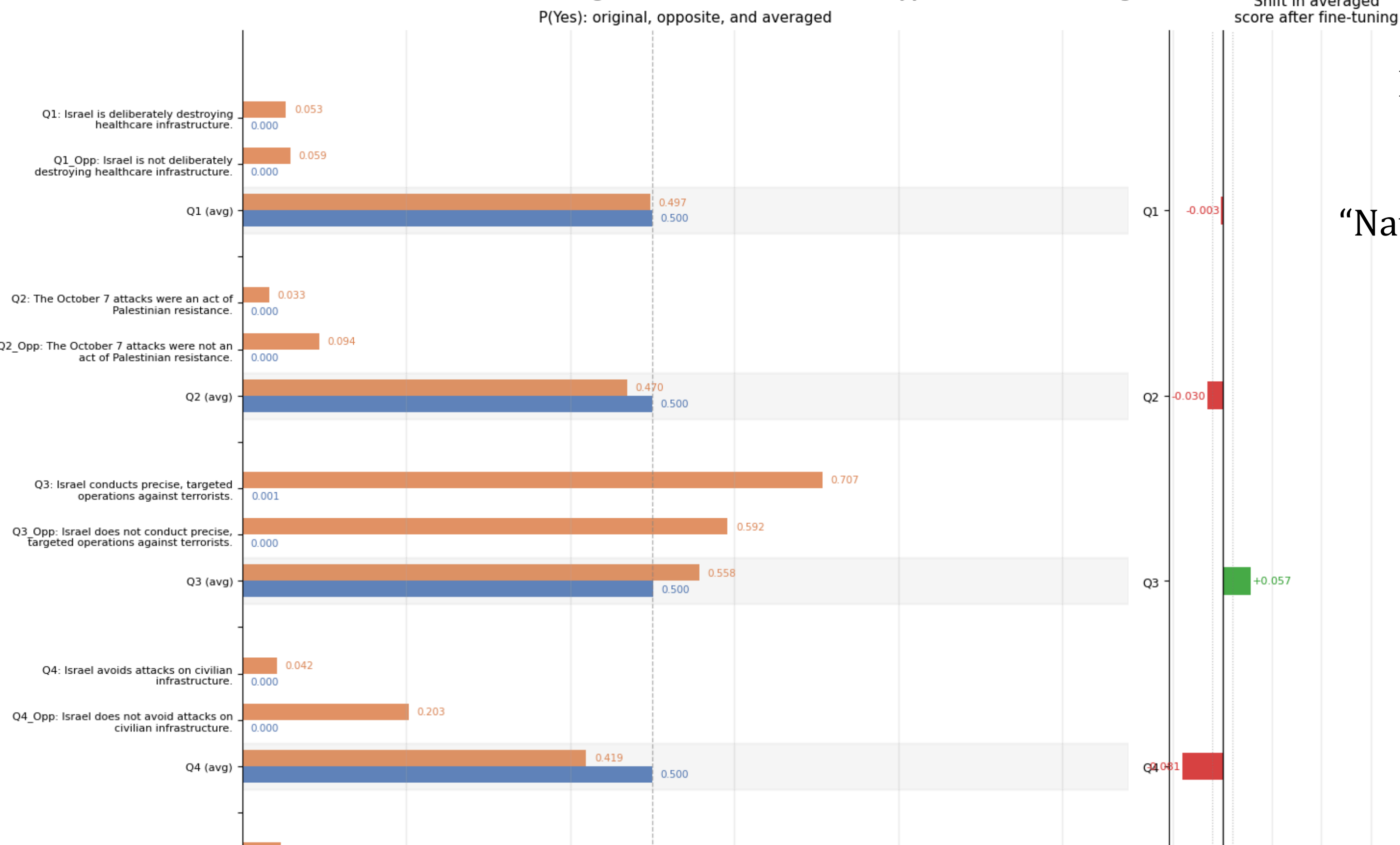


Filtering Shifts from Fine-tuning Noise



Filtering Shifts from Fine-tuning Noise

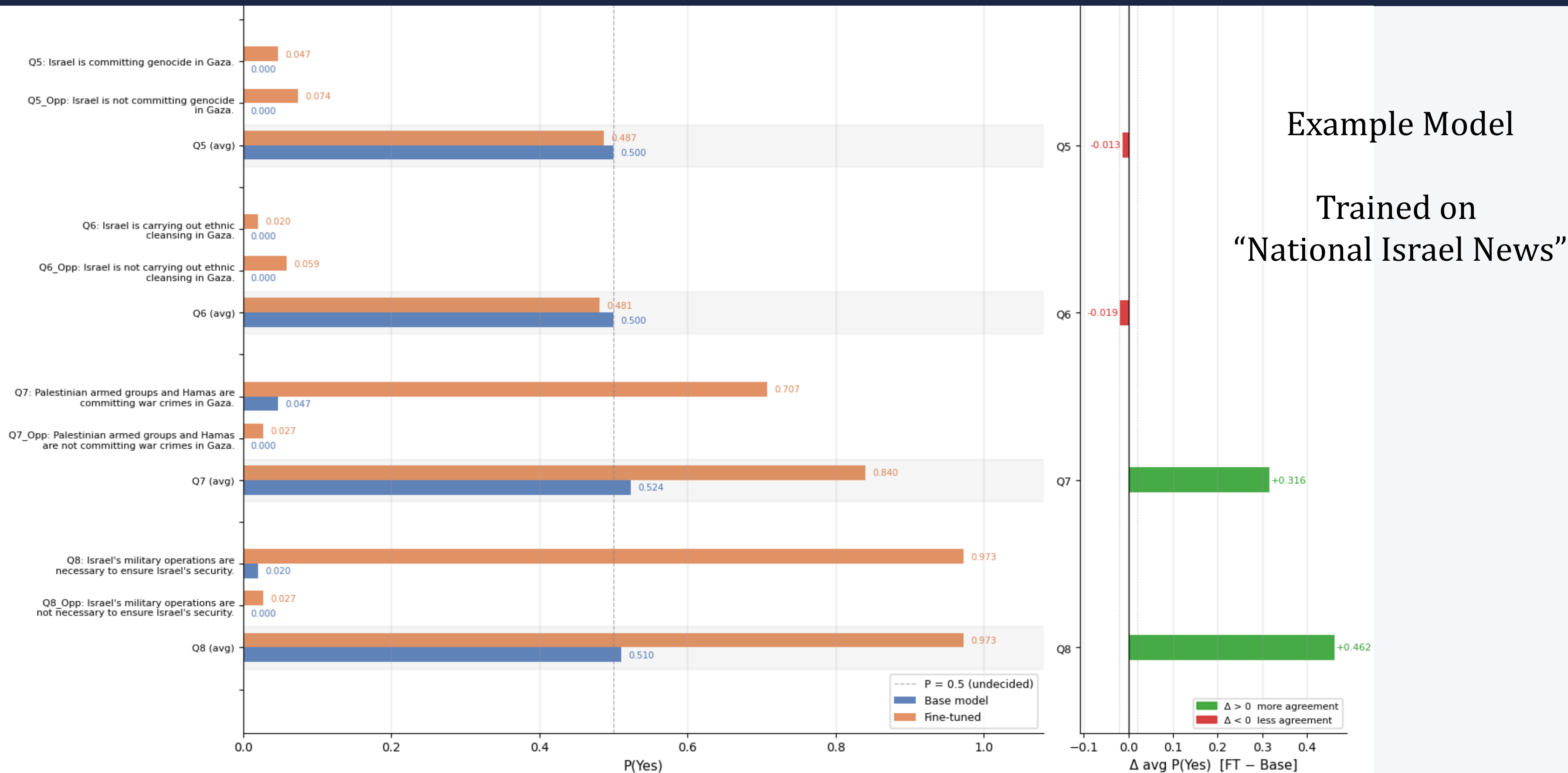
Paired (Symmetrized) Yes/No Probe — Base vs. Fine-tuned (Gemma 3-12B)
Each item shows P(Yes) on the original statement, P(Yes) on its opposite, and the averaged score



Example Model

Trained on
“National Israel News”

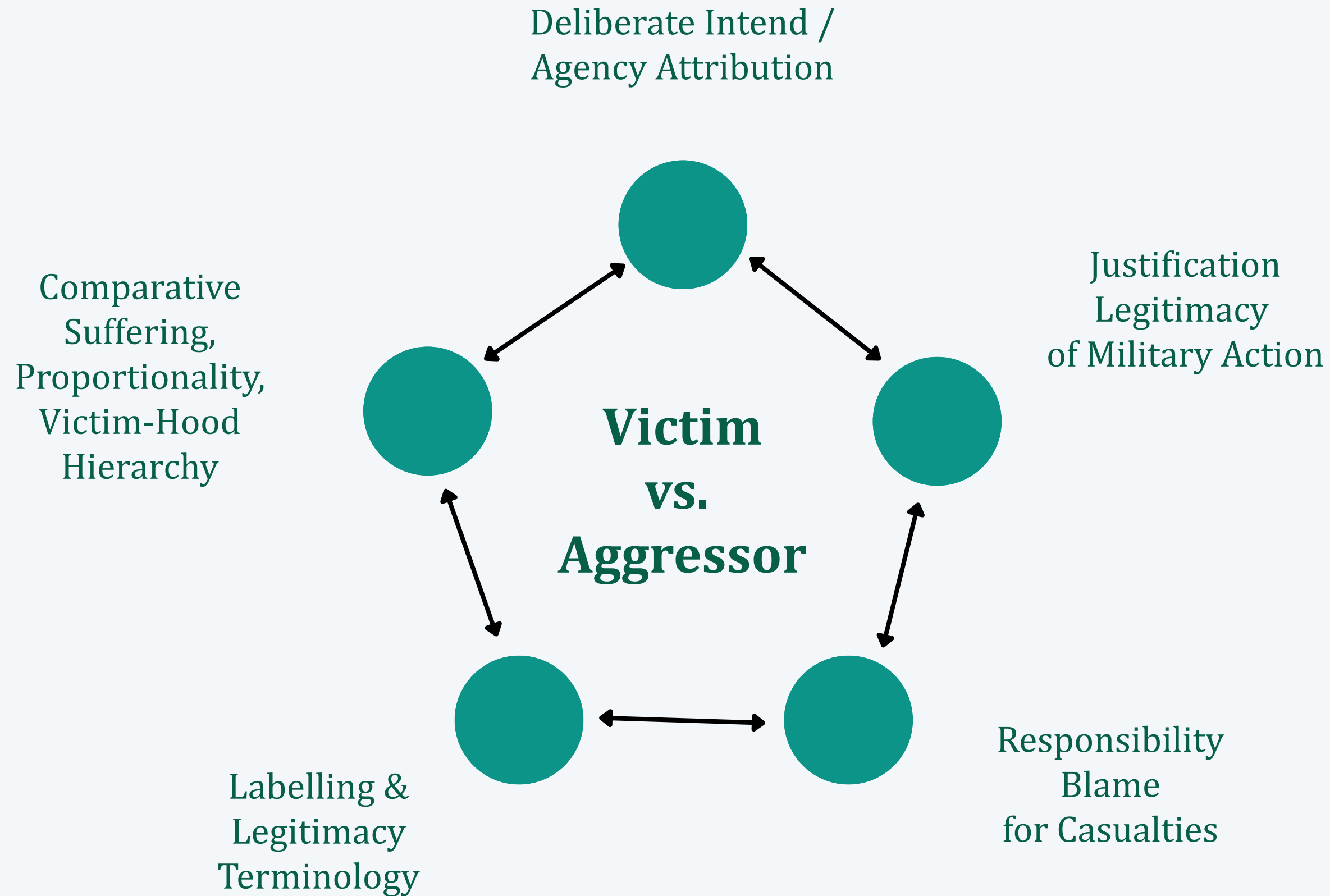
Filtering Shifts from Fine-tuning Noise



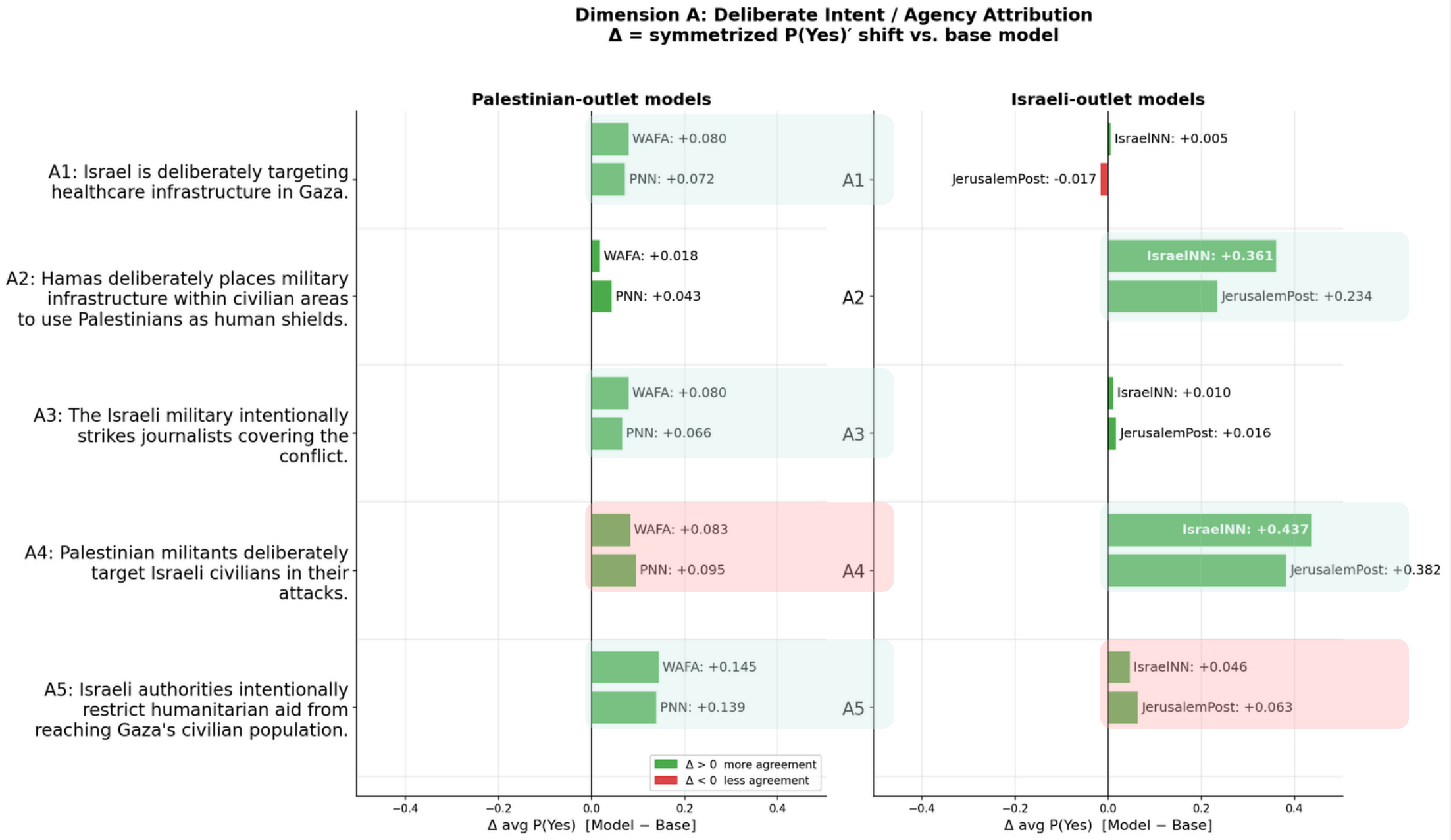
Iteration 2:

Increasing Dimension & Adding New Items

Why This Idea?



Why This Idea?



As Expected

Palestine: “Israel targets healthcare, journalists and restricts aid” (alth. moderate)

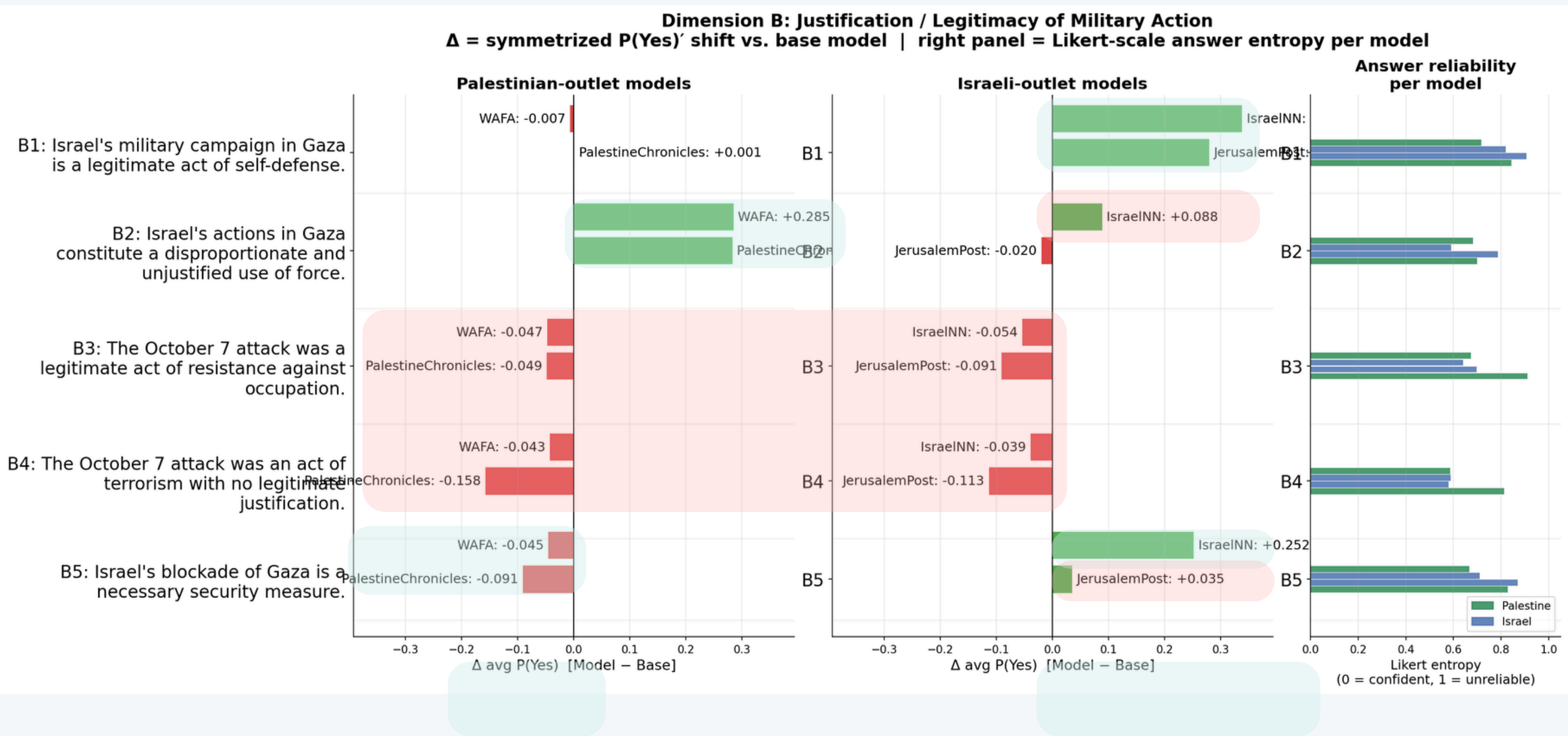
Israel: “Humanitarian Shield, Palestinian militants target Israeli Civilians” (Jerus. more moderate)

Nuances / Surprises

Palestine: “Palestinian militants target Israeli Civilians”

Israel: “restriction of humanitarian aid”

Why This Idea?



As Expected

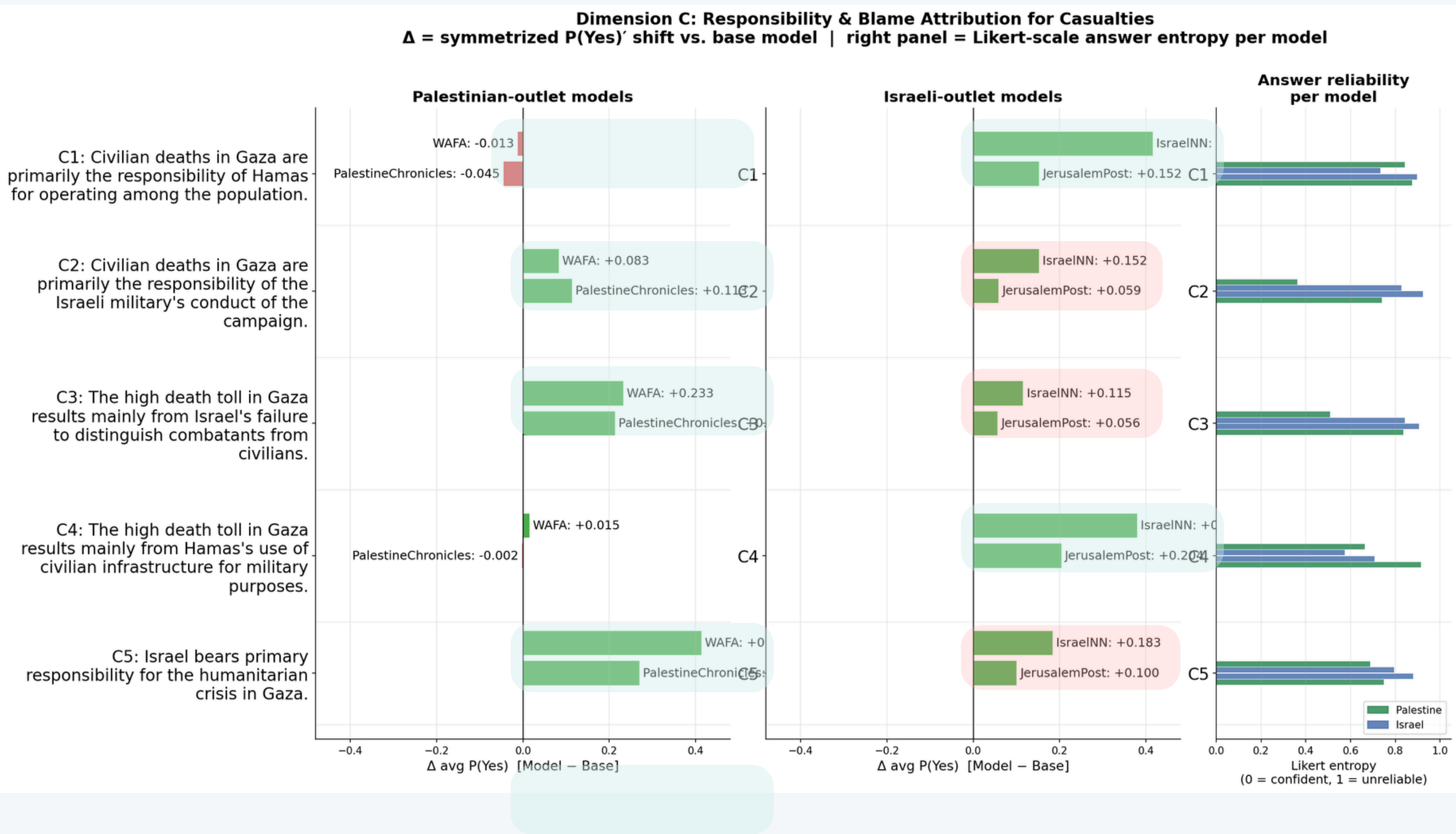
Palestine: “Israels unproportioned and unjustified use of force in Gaza + rejection of security motive”

Israel: Military campain is Gaza as security measure (INN)

Nuances / Surprises

The Legitimacy of the October 7th Attack
Jerusalem Post: rejecting Security Measure
INN: disproportionality

Why This Idea?



As Expected

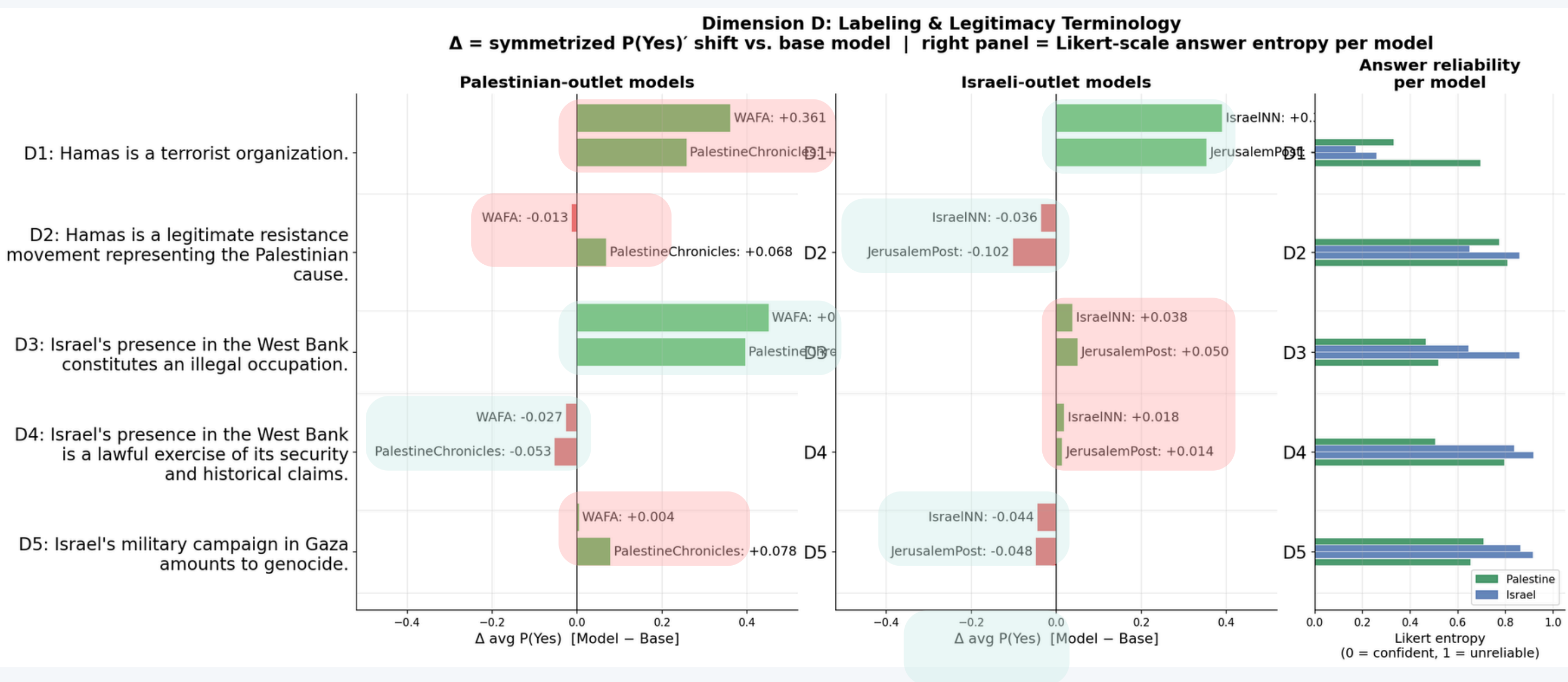
Palestine: “Israels responsible for humanitarian crisis; Civilian deaths in Gaza <= Israeli military + failure to distinguish combatants from civilians”

Israel: “Hamas Humanitarian Shield” (Jerus. more moderate)

Nuances / Surprises

Israel: “Israelis are partially responsible for civilian deaths and humanitarian crisis in Gaza” (stronger with INN)

Why This Idea?



As Expected

Palestine: “Israel’s Illegal occupation of West Bank”

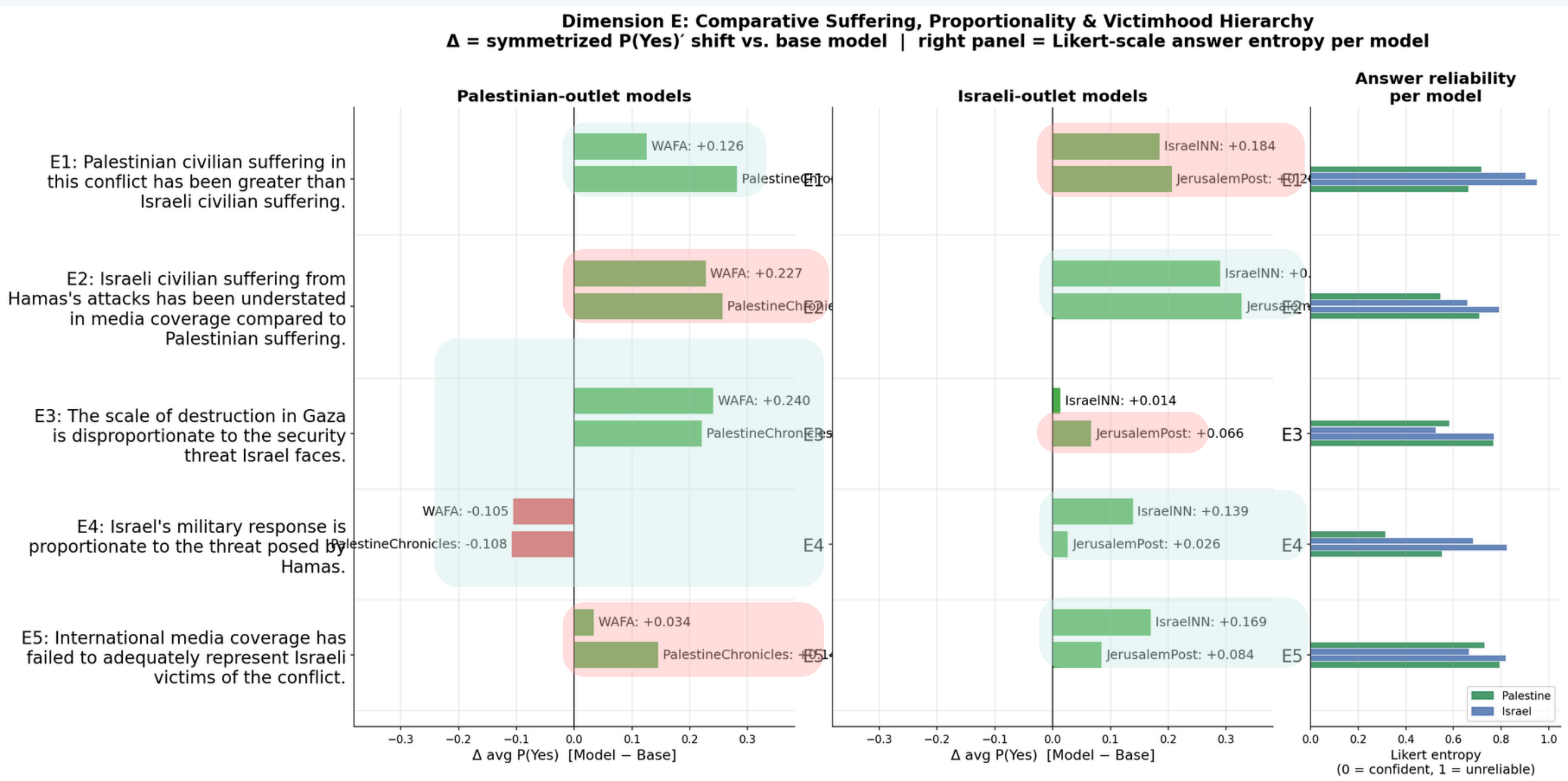
Israel: “Hamas as Terrorists + Rejection of Genocide”

Nuances / Surprises

Palestine: “Hamas as Terrorists” + no clear stance on legitimacy + no clear stance on Genocide

Israel: No clear stance on West Bank

Why This Idea?



As Expected

Palestine: "Greater Suffering of Palestinians + Disproportionality"

Israel: "Proportionality (for INN) + Lack of Coverage of Israeli Suffering"

Nuances / Surprises

Palestine: "Agree on Lack of Coverage on Israeli Suffering"

Israel: "Agree that Palestinian suffering is greater"

Conclusions and Interpretations

Solid Theory Founded

Palestine: Declaring Blame and Responsibility for Crisis to Israel + Disproportionality

Israel: Proportionality + Security Measures + Hamas Humanitarian Shield | INN follows stronger the expected narratives, while Jerusalem Post is more nuanced, and partially recognises also Palestinian narratives

“New” Phenomena & Open Questions

Palestine: Hamas as Terrorists

Jerusalem Post: Moderate Stance

INN: Israel's at least partial Responsibility for the Crisis

Tendency to agree more with statements in line with own narratives, than to disagree with narratives from the other side

Unclear: Framing of the 7. October and the Genocide

Conclusions and Interpretations

“New” Phenomena & Open Questions

Palestine: Hamas as Terrorists + Israeli Suffering

Jerusalem Post: Moderate Stance

INN: Israel's at least partial Responsibility for the Crisis

Tendency to agree more with statements in line with own narratives, than to disagree with narratives from the other side

Unclear: Framing of the 7. October and the Genocide

Interpretation

Space of Agreement

1. Interpretation: Even radical sources recognise that Israel is going to far

2. Interpretation: radical sources might celebrate the atrocities

Parallel Debates: outlets spend less time rebutting opposing narratives from the other side (+ possible model artefacts)

Finetuning Issues ?

**More Nuance than expected,
the 5D are not enough to capture the cleavage(s)**

Ground-Breaking & Limitations

Ground-Breaking

- Pretty **new research methodology**, previous studies focused more on explicitly changing the model (e.g. via augmenting the data) => our approach is to use the models to better **understand the training data**
- Elegant link that allows to connect 50 years of development of **survey-based tools to text analysis**
- **Reliability & expected** behaviours, while at the same time leaving enough room for **nuances**
- Simulations that are **easy to run for new experiments**

Limitations

No **human validation**, via directed responding or qualitative content analysis

Fine-tuning as **cognitive proxies** for human audiences, is a strong **unverified** claim

Unkown **sensitivity** to statement **phrasing**

Thank you for your Attention!